Q&A

Things You Wanted to Know!

Q1. Can I replace LEDs with a fluorescent tube in an emergency light project? If not, please explain in brief how fluorescent tubes work?

Jahnavi Lakshmi Purini

A1. Light emitting diodes (LEDs) and fluorescent lights use two different technologies for producing light. So, the short answer is no.

A fluorescent light is a type of gas discharge tube that contains inert gas within the glass casing while an LED is based on a PN junction diode, a solid-state device. Fluorescent tube produces ultraviolet (UV) radiation that is converted to the visible light. LEDs emit electromagnetic radiation across a small portion of the visible light spectrum.

Further, fluorescent tubelight requires a ballast circuit to stabilise the internal current that produces light while LED can be directly connected to a DC source through a current limiter to produce the light.

Fluorescent tube is coated with a layer of phosphor, which glows when it comes in contact with the UV radiation. Although most UV radiation stays within the tube, some does escape into the environment. This UV light can potentially be hazardous, while normal LED does not produce any hazardous UV light.

A fluorescent tube is almost always powered with AC. The tube's driver or ballast consists of a transformer along with other passive components. Transformer's primary coil current induces current in the secondary coil, producing a voltage of approximately 230V that makes the fluorescent coating to glow.

When power is first applied to the tube, a high voltage discharge is needed to start the flow of current. However, once this discharge takes place, a much lower voltage is sufficient—usually under 100V for tubes under 30 watts and 100V to 175V for tubes of 30 watts or more.

The fluorescent starter is a time-delay switch that opens after a second or two. During this time, the high voltage across the tube allows a stream of electrons to flow across the tube and ionise the mercury vapour, which makes the tube to glow. Some modern fluorescent tubes do not require a starter because they come with a ballast that has extra windings and energy to initiate the light.

Q2. If an SMPS for a PC is marked 400 or 450 watts, how is it calculated? How can I decide SMPS power requirement?

Anirvan

A2. Most switched mode power supplies (SMPS) for desktop PCs are AC to DC type. That means, SMPS takes AC mains input and outputs DC (3.3V to 12V). The DC outputs are used to drive various electronic circuits in the computer system including motherboard, hard disks, add-on cards, CPU, motors for fans, etc.

Power supplies are rated according to the maximum power (in watts) that they can produce. These parameters are normally specified by the manufacturers. Power output can be calculated if you know the rated output current and DC voltage output. For example, the SMPS specifications of a Traco power supply are listed in the table. Here, power outp ut = voltage × current = 24V × 18.75A = 450W.

It is important that your SMPS should be able to provide enough energy to drive all the devices in the PC at the same time. Most desktop PCs require power below 200 watts while running. But when you add external devices to your system, more power would be required. Also, more power is required initially when a system is switched on. Computer servers and workstations have lots of cards, drives, and adaptors, so power demand may increase beyond 350 watts, especially when the system is switched on.

In order to decide the SMPS power requirement, you need to consider the power of each device connected to the system and sum up their wattage to estimate the overall power requirement.

Traco Power SMPS Specifications

Brand	TRACOPOWER
Output current	18.75A
Number of outputs	1
Mounting type	Screw mount
Length	148.2mm
Depth	40.6mm
Maximum temperature	+80°C
Medical approved	Yes
Output voltage	24V DC
Power rating	450W
Input voltage	120 - 370V DC, 85 - 264V AC
Package type	Enclosed
Width	80mm
Weight	552gm
Minimum temperature	-40°C
Counrty of origin	TW

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